

Multimodal Sleep Stage Classification with LLM based Sleep Report

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Abstract—Manual scoring of polysomnography (PSG), the gold standard for sleep analysis, consumes time and is subjective in nature, hence presenting a major bottleneck in clinical workflows. This research tackles such bottlenecks by proposing a full scale end to end automated sleep stage classification and intelligent report generation pipeline. The system depends on multimodal data (EEG, EOG, EMG) from the public Sleep-EDF dataset and employs an advanced machine learning(ML) methodology. The pipeline begins with signal preprocessing, and then feature engineering takes place in all its impact, from both the time and frequency domain, ending with a novel multi metric feature selection approach aimed at resolving the paramount sleep biomarkers. Then, ML model classifies sleep into five different stages which are Wake, N1, N2, N3, and REM. The main contribution behind this work is by integrating a cutting edge Large Language Model (LLM) to generate the intelligible interpretations of quantitative classifier outputs, including parameters such as sleep efficiency, latencies, and stages.

Index Terms—PSG, Sleep Staging, Machine Learning, LLM.

I. INTRODUCTION

Sleep happens to be an important factor that concerns human health, and interrupted quality causes many chronic physiological and psychosocial illnesses [1], [2]. PSG is the clinical gold standard for evaluating sleep. This is an all night study that collects signals of multiple physiological modalities, including EEG for brain activity, EOG for eye movements, and EMG for muscular activities. The clinician's framework in analyzing these data is manual scoring by trained technicians or clinical experts, a process so time consuming and very costly, with plenty of variation from person to person, putting a huge tunnel of obstruction in the way of the practice of sleep medicine. To address these challenges, the automated sleep stage classification using ML has emerged as a better alternative, offering the potential for fast, efficient, and scalable analysis. The main objectives of this study is, to preprocessing multimodal PSG data. Then, to engineer and select a highly effective set of features from EEG, EOG, and EMG signals through a hybrid and novel feature selection technique for accurate sleep stage discrimination. And to train, validate, and compare ML pipelines for five stages of sleep (Wake, N1, N2, N3, REM).

Finally, to explore the use of LLM to automatically create a comprehensive and qualitative sleep analysis report from the model's predictions. Ultimately, the major aim of this study is to demonstrate the potential that such a system might be automated so as to improve efficiency and standardization in scoring sleep, along with facilitating accessibility of sleep health diagnostics through a novel LLM generated clinical summaries.

Sleep staging, the act of sorting 30 second PSG epochs into different sleep stages, has been a traditional manual scoring task that usually uses either the older Rechtschaffen and Kales (R&K) standard discussed in [3] or the more recent American Academy of Sleep Medicine (AASM) standard discussed in [4]. The AASM standard that is used in this work defines five stages which are Wake (W), N1, N2, N3, and REM. In the recent studies, deep learning (DL) methods such as CNNs [8], [9], [10], [11], RNNs [12], [13], transformers [14], and their hybrids [15], [16] have become more popular due to their capability of automatically deriving features for an end to end classification. As a result, the complexity of models has been growing along with the computational costs immensely such as SleepGPT [17] and DeepSleepNet-lite [18] for almost the same performance. These black box models do not always have the same level of interpretability as required in clinical adoption, and thus, they face some obstacles in being applied in the real world. Some researchers interpret the situation differently and claim that conventional ML presents a more feasible and interpretable option. Several studies emerge showing that a properly built pipeline, namely, deep, multi domain feature extraction from EEG and EOG signals along with efficient feature selection, is able to level with best in class DL implementations [19], [20], [21], [22]. For example, [23] one study combined a novel two phase feature selection scheme and used XGBoost classifier resulting in 87.0% accuracy. It was especially significant, as the DL models had difficulty in the N1 stage. Another study [24], adopted the ReliefF (ReF) method to select the most important features from EEG, EOG, and EMG signals

and then gave them as input to AdaBoost classifier. These outcomes point out to the fact that less complicated, more transparent models which are more user friendly in terms of training, deployment, and interpretation, are almost as good as those that are fully automated. The advantage of ML based classification and disadvantage of using DL models have been discussed in paper [25]. Hence, this research work decides to rely on manual feature extraction fused with the innovative feature selection method followed by classification with ML based techniques where accuracy as well as clinical adoptability are given the top most priority.

II. METHODOLOGY

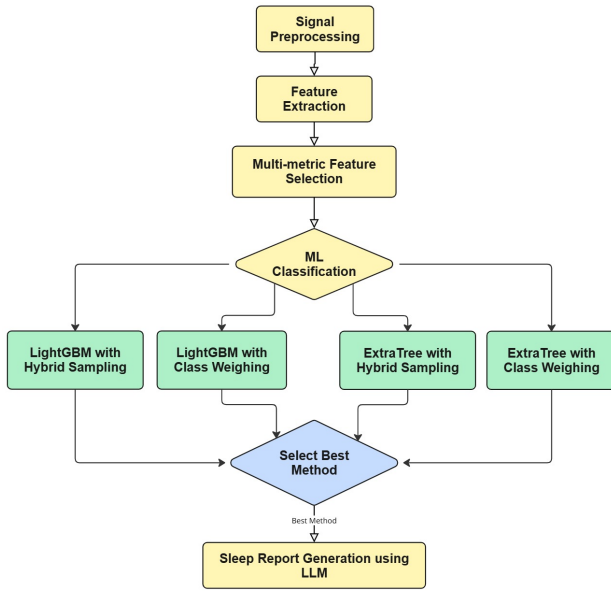


Fig. 1. Sleep Staging Pipeline

A. Dataset Overview

The Sleep-EDF Expanded Dataset [26] consists of whole night PSG recordings, totaling 197, of which each has corresponding annotations. The selected channels for this work are two EEG channels (Fpz-Cz and Pz-Oz), one horizontal EOG channel, and one submental EMG channel. The dataset contains two different studies which are Sleep Cassette (SC) and Sleep Telemetry (ST). The SC part consist of 153 healthy subject's recordings recorded at their homes to study the influence of age on sleep. The ST segment comprises of 44 recordings recorded from a hospital lab, which was structured to study the effect of a sleep aid on a group of individuals with mild difficulty falling asleep. The sleep stage annotations are based on the Rechtschaffen and Kales (R&K) manual and besides wake, N1, N2, N3, N4, and REM, it includes Movement stages as well.

B. Signal Pre-Processing

Preprocessing stage converts the original multi channel sleep data into a clean, standard format that is compatible with

ML. The procedure involves first loading each subject's PSG recording along with their corresponding hypnogram (sleep stage annotations). Each recording is adjusted to a standard frequency of 100 Hz by resampling. Wavelet denoising is being implemented as a part of preprocessing to clean the physiological signals, during which signal features are not compromised as shown in Fig 2 and 3. Traditional filters are less effective especially in the case of non-stationary signals like EEG, where the wavelet denoising technique excels [25]. There are three main steps involved in this technique and those are decomposition, thresholding, and reconstruction. Essentially the signal is decomposed into 7 frequency levels utilizing a wavelet function (db4 here). Thus the separation of the signal's basic components from the high-frequency noise is achieved. A noise threshold is statistically derived, and the signal components that are below the threshold are either reduced in size or set to zero in the so called soft thresholding operation. The final step is the reconstruction of the cleaned signal, i.e. the noise is removed but the important features are preserved for the correct sleep stage classification.

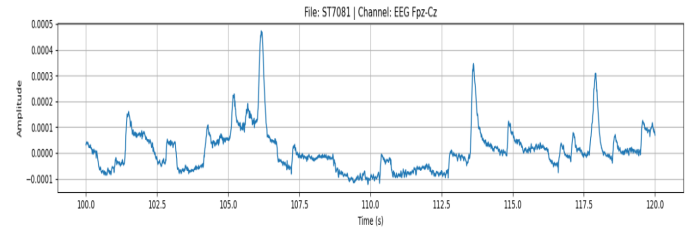


Fig. 2. Before Preprocessing

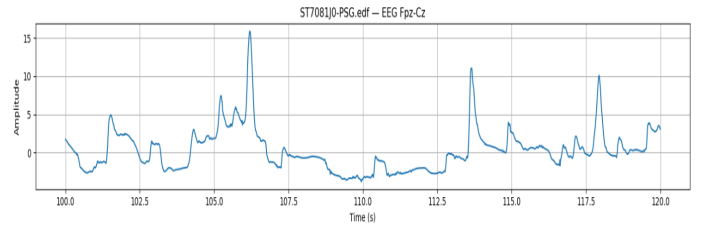


Fig. 3. After Preprocessing

The continuous part of the data is then cut into 30-second non overlapping epochs after noise removal which is the standard for sleep stage scoring [4]. Then a per subject, per channel Z-score normalization is applied, wherein the mean and standard deviation for each channel are calculated from the data of one individual's file only, and this is then used to scale the data. The main advantage of this method is that it effectively gets rid of the huge, most of the time irrelevant variations in signal amplitude that happen naturally between different people. Consequently, the ML model becomes capable of detecting genuine foundational patterns in the data instead of person specific baselines, thereby its capacity to generalize to new unseen subjects is enhanced considerably.

C. Feature Extraction and Selection

Feature extraction had a significant impact on the sleep data representation, as it managed to combine the standard metrics with several novel and advanced features that can capture complex signal dynamics. Alongside the typical time domain (e.g. mean, variance, RMS) and frequency domain features (e.g. absolute and relative band powers), the analysis has also included more advanced measures. The Hjorth parameters (mobility and complexity) were features which in a very efficient manner, represent the statistical and frequency characteristics of the EEG signal. Permutation Entropy was set as a impactful, non linear measure to outline the complexity, and predictability of EEG, thereby differentiating the various stages of brain activity. Teager-Kaiser Energy Operator (TKEO) was deployed on EOG and EMG signals for detecting transient events to calculate the instantaneous energy, which, in turn, made it very successful in locating the eye movement bursts along with the muscle twitches that are the characteristics of certain sleep stages. This multi faceted approach not only ensures that the model is getting trained on the features that depict the basic signal properties but also their complex dynamic behaviors.

The feature selection step used here is a novel heterogeneous operation that aimed to judiciously trim down the original 99 features to a resilient and effective subset of 67. The stage did not depend on one metric only, but went on to use a 5 fold cross validation as an advanced ensemble strategy which takes into account of multiple metrics to reach a decision. For each feature, three different ways were used to calculate the value which are the gain of LightGBM, mutual information, and permutation importance. Then these raw scores were reprocessed by adding a stability score that accounted for the features which were ranked highly and thus in a stable manner across all the validation folds. Moreover, based on [4], the knowledge of the domain was included through the use of modality weights that gave priority to the features of EEG signals than those of EOG and EMG. The weights are as follows EEG: 0.5, EOG: 0.3 and EMG: 0.2. They were ranked by combining these metrics into one final score and then the top 80 features were picked. In the last stage, this catalog was cleansed using correlation filtering of highly correlated features (those with a correlation over 0.9) to assure that the final set was not only highly predictive but also minimally redundant.

D. Machine Learning

The class distribution of 5 classes is shown in Fig 4. Class 0,1,2,3 and 4 signifies to Wake, N1, N2, N3 and REM. The ML phase was a thorough comparative process which aimed to discover the best model to classify sleep stages by using the features created during the process. Two ensemble models, LightGBM and ExtraTrees Classifier, which are both strong, were systematically evaluated. As there was a significant imbalance between the classes in the dataset as shown in Fig 4, with the stage of wakefulness being the most frequent, two

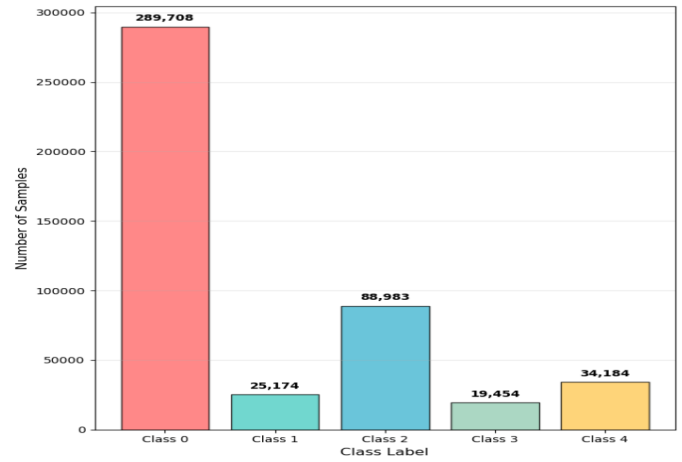


Fig. 4. Class Distribution

different approaches were tested for each model. Firstly, the use of class weighting to penalize the errors of the minority classes more heavily and secondly, the application of a hybrid resampling pipeline that combined SMOTE for synthesizing data for the rare stages with Random Under Sampler(RUS) for reducing the majority class. The whole experiment was done with 5 fold stratified cross validation to obtain a reliable and robust evaluation. The SMOTE with RUS was applied only to the training folds within cross validation using an imblearn pipeline, ensuring no data leakage from test folds. By looking at the performance of the models using different metrics and focusing mainly on the macro F1 score and the overfitting gap between the train and test scores, it was decided that the LightGBM model combined with the SMOTE and RUS pipeline is the best way to go, thus, it got the best balance of high predictive accuracy and strong generalization. This last, optimized model was then retrained on the entire dataset and stored for the final inference and report generation pipeline.

E. Sleep Report Generation

One important transformation in the sleep report generation step is the rendering of the raw, numerical predictions of the ML model into a detailed, understandable analysis for a human user by an LLM. The method starts with producing a quantitative summary from the sleep stages that have been predicted. This summary contains key clinical metrics such as Total Sleep Time, Sleep Efficiency, Sleep Latency, and the percentage of time spent in each stage (Light, Deep, and REM) calculated based on AASM sleep scoring data [4]. The data in this summary, which is organized, is then used as the basis for a prompt that has been designed with the Gemini LLM in mind. The LLM used for report generation is gemini 1.5 flash. Gemini 1.5 flash was chosen as it offers ideal balance of performance, speed, and cost-effectiveness without compromising quality. In fact, this model is very efficient in capturing the structured sleep data and also in understanding the prompt which is quite complex as well. Being a flash type, it achieves these outputs very quickly with almost zero waiting

time and at a lower cost, which is perfect for large volume applications that are quality and efficiency demanding. The prompt contains guidance for the model to take the role of a sleep research specialist and create an in depth report with certain parts, such as overall summary, Sleep Score out of 100, detailed analysis of sleep architecture with reference to normal ranges and feasible recommendations. The sleep score is calculated by the LLM based on the sleep metrics. The evaluation integrated within this process is very instrumental whereby the quality of the generated report is checked in two forms i.e a factual consistency check by a separate judge LLM to avoid numerical mistakes, and an assessment of semantic similarity with the use of BertScore between the generated and ground truth summary. This allows the final output to be not only fluent and nicely formatted but also accurate and dependable. BertScore attempts to semantic align a candidate sentence with a reference sentence. The paper [27], explains how BertScore is calculated.

III. RESULTS

A. Baseline Comparison

The detailed results from the four experimental conditions show beyond a doubt that a multi modal model that combines all features (EEG, EOG, and EMG) is decisively superior for automatic sleep stage classification. The full model obtained an overall accuracy of 90.3% and a macro average F1-score of 0.82. This is a big jump over the next best model (EEG-only, 0.75 F1-score), indicating that the signal combination is not just additive but synergistic, thus allowing the model to capture the complex physiological states that characterize sleep architecture more accurately than any single signal type can by itself. The EEG only model can be regarded as the most important single variable in the prediction of sleep stages and it can set a benchmark for the multi modal model on its own with a macro F1-score of 0.75. This is the expected result since the electroencephalogram is a direct measurement of brainwave patterns that are the main criteria for staging sleep, such as the slow delta waves of N3 and the sleep spindles of N2. Where the EEG only model is most confused is between stages that have similar low voltage, mixed frequency brainwave patterns. The same detailed results reveal this in the treatment of REM sleep, where the performance of the EEG only model is quite low with an F1-score of 0.68, as it often mistakes REM for N1 or N2 sleep.

The main drawback of the multi modal approach is fixed in such a way that EOG and EMG signals are EEG dependent. The most prominent case is the REM sleep detection. The F1 score for REM increased sharply from 0.68 in the EEG only model to 0.83 in the full model. This is because the EOG features pick up the rapid eye movements while the EMG features pick up the muscle atonia, which are the two main non EEG markers for REM sleep, thus allowing the full model to make the most of these three signals for reliable predictions of N1 sleep stage and the highest F1 score (0.55) for this stage. Last of all, the EMG only model is an example

of the worst standalone predictor, with a macro F1 score of only 0.44 and almost complete failure to identify N2 sleep ($F1=0.26$). It is clear that muscle tone alone cannot be relied on for total sleep staging. Its indispensable contribution to our identification of muscle atonia during REM sleep and high muscle tone during wakefulness is the one that enables the better performance of the multi modal model to happen. The detailed comparison of results from each feature sets is shown in Fig 5.

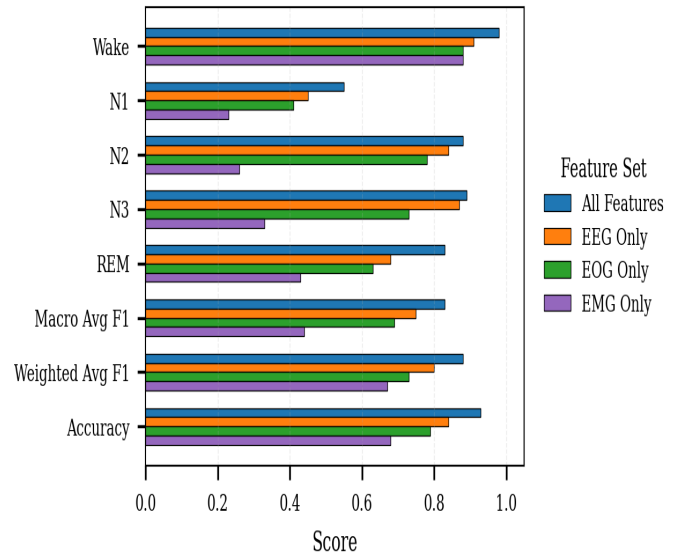


Fig. 5. Performance Metrics for Different Feature Sets

B. Results of the Proposed ML Framework

The output explains very well that the most efficient strategy to model the data. The results shows that the LightGBM models are always better and significantly so, over the ExtraTrees models regardless of the metric used. It is, therefore, no wonder that the Gradient Boosting algorithm, which sequentially builds trees that correct the errors of the previous ones, is more successful with this complex, tabular features set than the bagging approach of ExtraTrees. When we look at each model in detail, the primary distinguishing point is how the resampling has been handled either SMOTE along with RUS or class weighting. Essentially, this decision affects only a minor yet steady influence on the general performance of the model, suggesting that the two methods are roughly equally effective with minor differences in the stability. For example, in the case of the best LightGBM model, the F1 score on the test set was moved from 0.7835 to 0.7935 by SMOTE with RUS. This signifies that the combination of generating synthetic data for the minority class (N1) and down sampling the majority class (Wake) results in a more balanced and efficient training environment for the model. Hence, the LightGBM with hybrid sampling(SMOTE with RUS) pipeline is the best performing model without, showing

the most successful combination of a potent algorithm and a clever class imbalance handling technique. The performance comparison is shown in Table I.

TABLE I
COMPARISON OF MODEL PIPELINES AND THEIR TEST SCORES

Model Pipeline	Metric	Test Score
LightGBM with SMOTE and RUS	Accuracy	0.9034
	Macro F1-Score	0.7935
	Cohen's Kappa	0.8263
LightGBM with Weighted balance	Accuracy	0.8876
	Macro F1-Score	0.7835
	Cohen's Kappa	0.8036
ExtraTrees with SMOTE and RUS	Accuracy	0.8202
	Macro F1-Score	0.6791
	Cohen's Kappa	0.6937
ExtraTrees with Weighted balance	Accuracy	0.8155
	Macro F1-Score	0.6760
	Cohen's Kappa	0.6880

C. Best Model's Strength and Weakness

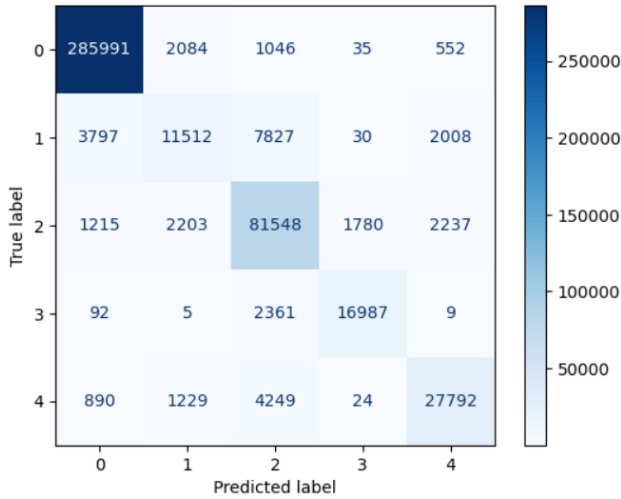


Fig. 6. Confusion Matrix

The LightGBM with hybrid sampling method is strong in various aspects, particularly the generalization capability and the accuracy on the most typical sleep stages. The model's consistency can be interpreted from the training and validation loss charts, which follow each other very closely before leveling out. A stable model is indicated by this situation as it is far from overfitting. The tiny overfitting gap of only 0.0245 in the F1-score is thus in line with the argument of reliability of the model's performance and the possibility to generalize well to new data. In addition, the model is very effective in making predictions for stable physiological conditions, as it demonstrates this aspect by achieving the top F1-score of 0.98 for Wake (Stage 0) and very good results of 0.88 for N2, 0.89 for N3 (Deep Sleep), and 0.83 for REM (Stage 4). Nevertheless, the first and foremost model weakness is the complicated and incorrect identification of the transitional stage of N1 sleep (Stage 1) as seen in Fig 6, for which it just got an F1-score of 0.55. By examining the confusion matrix,

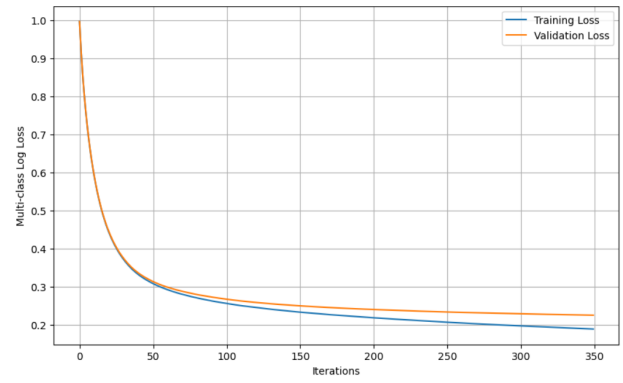


Fig. 7. Loss Curve

the reason for this limitation can be spotted, i.e., large numbers of true N1 epochs have been categorized as N2 sleep or Wakefulness with 1048 epochs and 1,215 epochs, respectively. This kind of mixing is a common and unproblematic case in sleep research, as N1 is a very brief and indistinct state between being awake and going to sleep. Still, the improvement in the classification of this specific stage is the biggest area of the model's future work.

D. LLM Response

The LLM assisted documentation system exhibited the capability to combine the raw data of a new sleep recording into a coherent, accurate, and clinically relevant narrative. It was prompted by inputting the last sleep metrics like total sleep time of 483 minutes and sleep efficiency of 34.1% into a structured prompt for the Gemini 1.5 Flash model. Consequently, the model dutifully carried out its orders, realizing a multi part report with an overall summary, a sleep score with justification, a breakdown of the sleep architecture, and recommendations for future actions. The report that was generated at the end of the day is very much readable and also it is in line with the data such as it is identifying the extremely low sleep efficiency and high sleep latency as the most critical issues which is a key indication of advanced, context aware reasoning. First, the new recording was passed as input to the pipeline and second, there was a verification of the accuracy of the report generated. A two pronged evaluation framework was used for this purpose, which gave the green light to its high quality. The semantic similarity between the generated sentence and reference sentence, powered by BertScore [27], produced an F1-score of 0.7837. This indicates a high degree of semantic overlap between the LLM's report and the ground-truth reference summary. Accordingly, the model was able to retell the text with the right clinical meaning and the main insights. Besides that, the "LLM as a Judge" evaluation for factual consistency gave the report a score of 4 out of 5. In the judge's opinion, the report was largely accurate, with only one percentage likely a typographical error. This means the LLM did not hallucinate or make up data, but instead, it based its account of events on the given metrics, and hence, its output became both reliable and trustworthy.

IV. CONCLUSION

This research has successfully took the responsibility to build and prove the validity of a complete working pipeline that allows for automatic classification of the stages of sleep and the generation of an intelligent report by means of the Sleep-EDF Expanded Dataset. It is through this task of data preprocessing, extensive feature engineering, and the implementation of the multi criteria feature selection strategy that the training of a LightGBM classifier was conducted to reach a high macro F1-score of 0.79. This result shows the classifier's potent ability to correctly identify sleep stage from the complex physiological signals. The most novel element of the present work is the transition stage in which outputs from the ML model were converted to visual, non-technical, and intuitive human output by Gemini's LLM. The accomplishment of this synthetic step has been quite thoroughly confirmed through a double evaluation framework, which indicated that, when compared to factual references, high factual consistency and semantic relevance were exhibited. One great thing about this kind of study is that it has the possibility to make the whole sleep field new and better. By employing ML to forecast and LLMs to interpret, this system would not only enable the clinician to do his work faster but also provide the patients with uncomplicated and easily understandable information about their sleep pattern. The feature based model implemented revealed as very efficient; the authors might consider using end to end DL strategies that could result in a better performance deriving intricate pattern from raw signals. In this regard, this study shall also be complimented on its promise of future clinical testing on different datasets. This will further facilitate its application in the real world. In general, this undertaking is an excellent illustration of the collaboration pattern between the classics of ML and the best in class generative AI paradigm in the progress of personalized medicine.

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